CSE 303 STATISTICS FOR DATASCIENCE

MINI PROJECT

VERIFIED-PURCHASE PREDICTION AND REVIEW TRUSTWORTHINESS

Core Focus:

Can linguistic and behavioral signals distinguish verified from non-verified Bangla book reviews on Rokomari.com?

Presented to
Dr. Nafis Sadeq, Assistant Professor,
Dept. of Computer Science and Engineering

AUGUST 15 2026

SECTION 6

Group Members

NAME	ID
Mahadi Prodhan	2023-2-60-033
Rishad Imtinan	2023-3-60-014
Md Mahiruddin Chowdhury	2023-3-60-504

Ethical Framing Choice & Problem Formulation

Rokomari relies on reviews, but unverified reviews may affect buyer trust.

Unverified reviews are not automatically fake so they may come from real readers who bought books offline or from libraries.

Predict purchase verification and analyze review trustworthiness.

Binary classification using Logistic Regression to predict $P(y=1|x)$.

Selected Variables, Target Label & Preprocessing Steps

Target Label Selection

Evaluated 3 candidates. Rejected Star Ratings (>80% 5-stars) and Helpfulness Votes (>90% zero votes). Selected verified_purchase (y in {0, 1}) as the objective target label.

Candidate Target	Data Structure & Limitations	Selection Decision & Rationale
Star Ratings (1 to 5 Stars)	Over 80% of reviews were 5 stars	Rejected: Extreme positive skew masks model learning.
Helpfulness Vote Ratio	Over 90% of reviews had zero votes	Rejected: High voting sparsity invalidates continuous regression.
Verified Purchase Status	Clean 50/50 split (verified vs unverified)	SELECTED: Balanced, objective ground-truth target label.

Table 1: Candidate Target Variable Evaluation & Selection Decisions.

Selected Variables, Target Label & Preprocessing Steps

Input Features/ Variables

6 Continuous Features: Word count, character count, punctuation intensity, helpfulness ratio, rating deviation, book price.

13 Categorical Features: One-Hot encoded top categories (dropped reference category "Collections Box Sets")

Total 20 Features+ Bias

Preprocessing

Removed 538,386 duplicate review pairs (72.33%) to satisfy statistical independence.

Min Max Normalization where bounded continous features to [0.1]

Log Scaling was done that is, log(1+x) to compress heavy right skewed text lengths

Determining the Right Sample Size

- Using all 400,000+ rows introduces noise, duplicates, and high computational cost without significant accuracy gains.

- Multi Trial Sampling Experiment:

  Trial 1 (N = 1,000): High word count Standard Error (SE = 4.4247).

  Trial 2 (N = 20,000 Stratified): Standard Error dropped by 77.5% down to SE = 0.9936.

- Split of 10,000 / 10,000 used as CLT supports a balanced 20k sample, giving tighter confidence intervals, 50/50 balance, and faster training convergence.

Rating Bias & Log-Scale Transformation

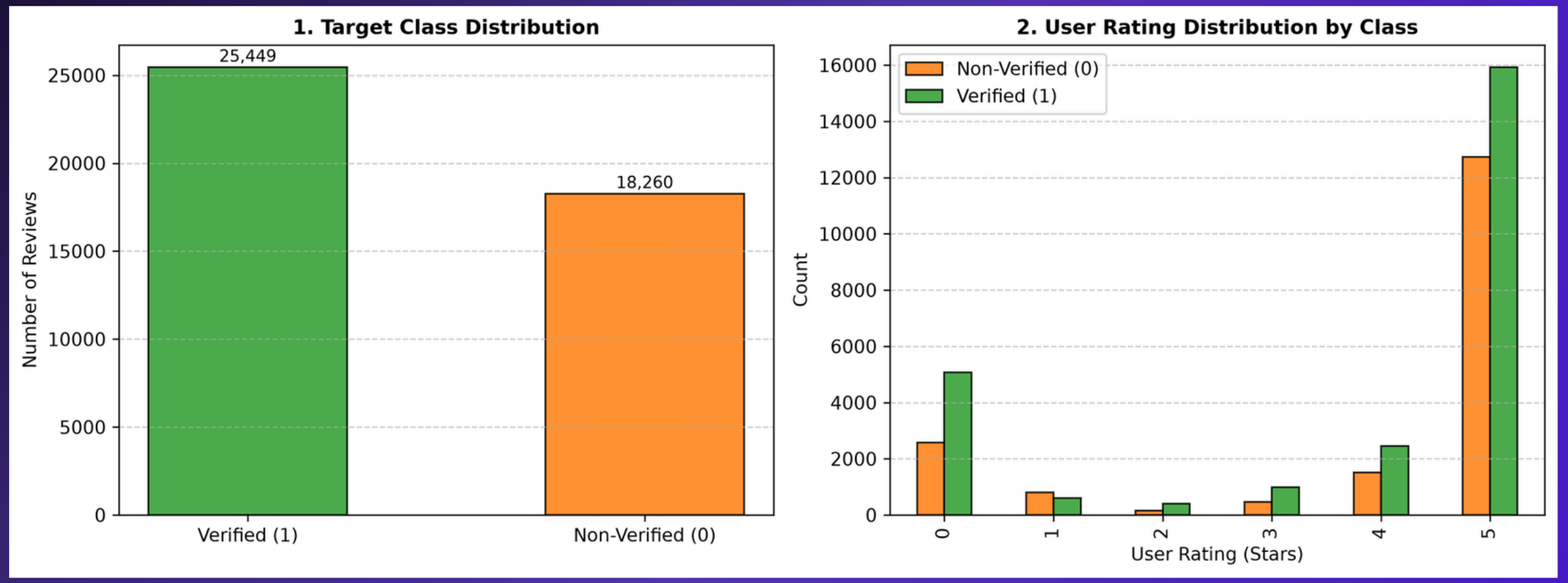


Figure 1: Target Class Distribution & Star Ratings Breakdown.

- Class Distribution & Rating Bias gives final analytical subset which contains 25,449 verified (58.22%) and 18,260 non-verified (41.78%) reviews.
- Non-verified reviewers leave 5-star ratings at a significantly higher rate (69.71%) than verified buyers (62.55%).

Rating Bias & Log-Scale Transformation

- Histogram Sensitivity Trial: Tested 15, 30, and 60 bins. 30 bins gave the optimal resolution without over-smoothing or visual noise.

- Logarithmic Scaling log(1+x): Raw review lengths range from 1 to 2,546 words. Applying log(1+x) compressed extreme right skew into a symmetric bell curve.

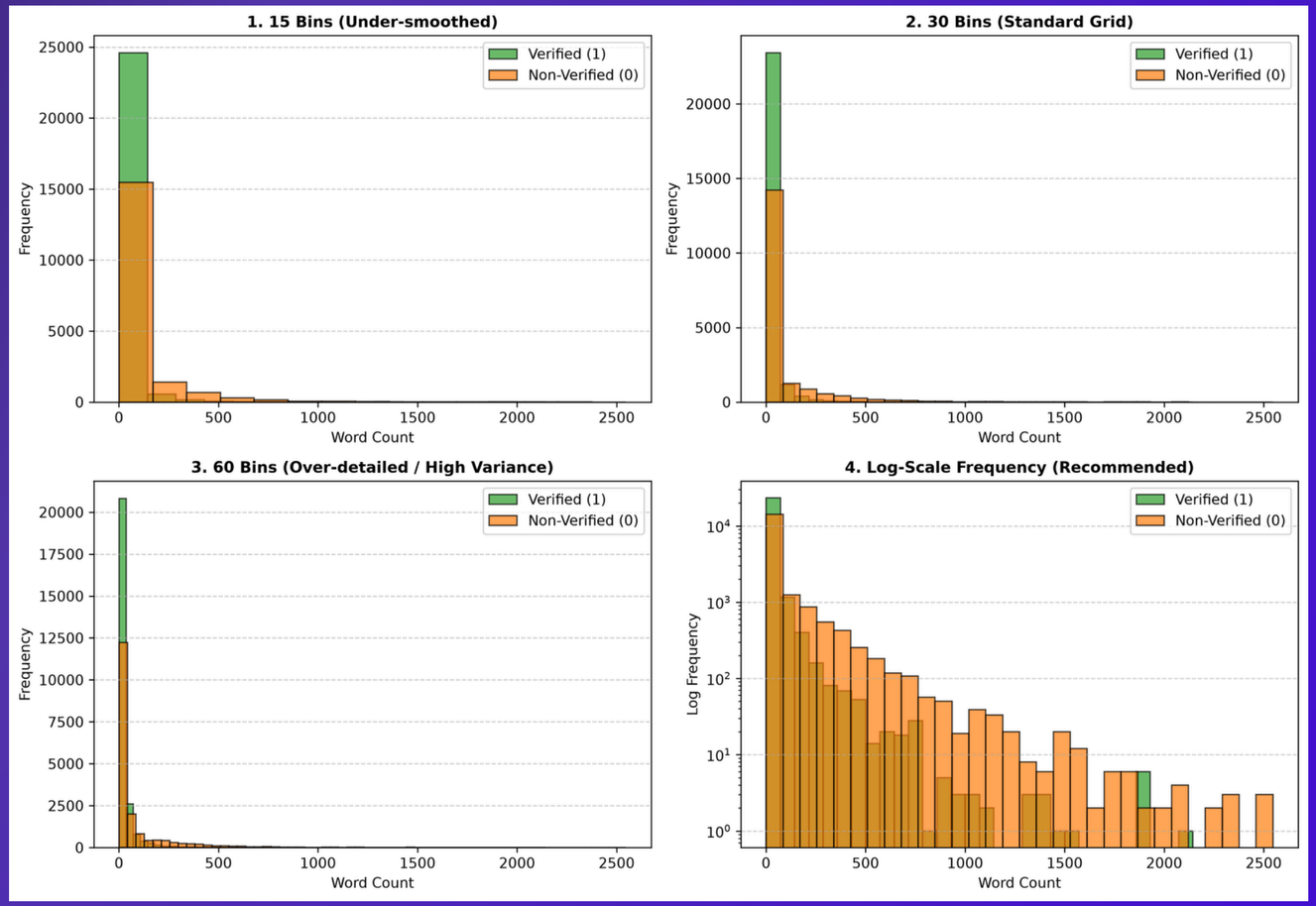


Figure 3: Raw Linear Scale vs. Log-Scale Transformation.

Outlier Robustness

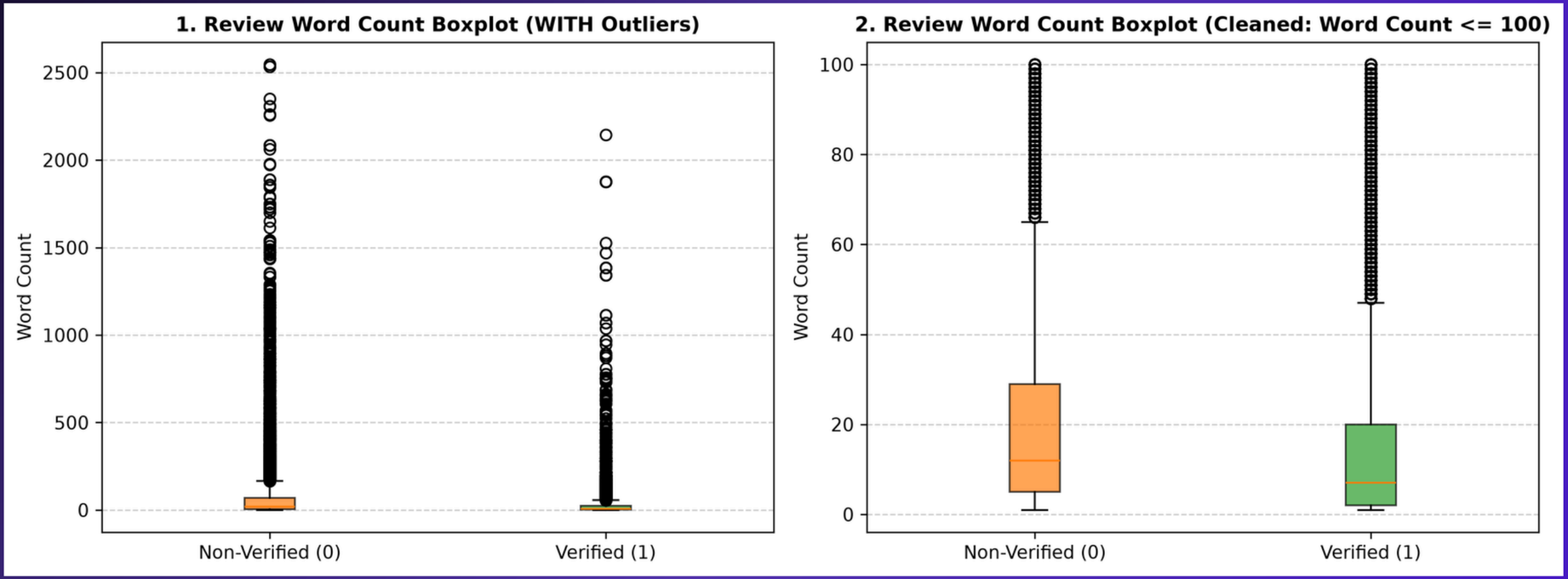


Figure 4: Review Word Count Boxplots WITH vs. WITHOUT Outliers.

- Outlier Analysis (Cleaned <= 100 words): Non-verified mean word count dropped from 89.78 to 21.12 words (-68.66 shift) and variance fell by 98.6%.

Outlier Robustness

Statistical Metric	With Outliers (Raw)	Without Outliers (Cleaned <= 100)
Non-Verified Mean Word Count	89.78 words	21.12 words (-68.66 shift)
Verified Mean Word Count	28.02 words	15.56 words (-12.46 shift)
Non-Verified Median Word Count	19.00 words	12.00 words
Verified Median Word Count	8.00 words	7.00 words
Non-Verified Variance	37,020.46	525.78 (-98.6% drop)
Verified Variance	5,755.09	400.07 (-93.0% drop)
Non-Verified IQR	64.00 words (Q1: 6, Q3: 70)	24.00 words (Q1: 5, Q3: 29)
Verified IQR	22.00 words (Q1: 2, Q3: 24)	18.00 words (Q1: 2, Q3: 20)

Table 2: Statistical Metrics Comparison With vs. Without Outliers.

Statistical Insight: Sample Mean is vulnerable to outliers, while Sample Median is robust (remained stable at 19.0 vs 12.0 words).

Parametric Hypothesis Testing

Z-Test and T-Test

3 Parametric Hypothesis Test with alpha = 0.05 was conducted

Results Were:

Z-Test (5-Stars): Z = -15.53, p = 0.0000 -> Significant rating bias.

Welch's T-Test (Word Count): T = -41.15, p = 0.0000 -> Non-verified reviews are significantly longer.

Metric	Value
Verified (1) 5-Star Count	15,919 / 25,449 (62.55%)
Non-Verified (0) 5-Star Count	12,729 / 18,260 (69.71%)
Pooled Proportion	0.6554
Standard Error (SE)	0.004609
Z-Statistic	-15.5289
p-value	< 0.001
Conclusion	Reject H_0 at $\alpha = 0.05$

Two Sample Z Test Results

Metric	Value
Verified Mean Word Count (u1)	28.02 (std: 75.86)
Non-Verified Mean Word Count (u2)	89.78 (std: 192.41)
T-statistic	-41.1466
p-value	0
Conclusion	Reject H_0 at $\alpha = 0.05$. Non-verified reviews are significantly longer

T-Test Results

Parametric Hypothesis Testing

One-Way ANOVA

Results Were:

One-Way ANOVA (Categories): F = 115.94, p < 0.0001 -> Significant genre differences.

Running 10 pairwise t-tests inflates Type I error to >40% (1 - 0.95^10). One-Way ANOVA keeps error controlled at 5%.

Metric	Value
Calculated F-Statistic	115.9362
p-value	0
Between-Group SS (SS_between)	43.2539 (df1 = 4)
Within-Group SS (SS_within)	1055.4539 (df2 = 11316)
Mean Square Between (MS_between)	10.8135
Mean Square Within (MS_within)	0.0933
Conclusion	Reject H_0 at $\alpha = 0.05$. Helpfulness ratio varies significantly across book categories.

One Way ANOVA Results

Feature Selection

Selected Feature Set

20 Input Features and 1 Bias

6 Continuous Metrrics

Word count, character count, punctuation intensity, helpfulness ratio, rating deviation, book price.

13 Categorical Columns

One-Hot encoded top categories (reference category dropped to prevent multi-collinearity)

Two Different Sets

Train-Validation -Test Partition was done on 2 different sets to choose the best

Train-Validation-Test Partition Procedure (3-Way Stratified Split)

Raw Data

20 Features+ Bias

✅ Set1

70% Train(30596)

15% Val (6,556)

15% Test (6,557)

❌ Set2

85% Train(37152)

5% Val (2185)

10% Test (4372)

70/15/15 Split over 85/5/10? An 85/5/10 split leaves only ~2,185 validation samples, creating noisy hyperparameter evaluations. The 70/15/15 split maintains 6,556 validation samples for stable Grid Search tuning

Models

5 Models Built & Compared

Model 1

Constant Mean Predictor
Predicts the same mean probability (0.5822) for all observations. It achieved 58.22% validation accuracy with BCE loss = 0.679567.

Model 2

Constant Median Predictor
Predicts the same median class (1) for all observations. It achieved 58.22% validation accuracy.

Model 3

L2 (Ridge) Logistic Regression
Custom NumPy gradient descent with BCE loss and L2 regularization to smoothly reduce weight values. Loss decreased from 0.693150 to 0.662313.

Model 4

L1 (Lasso) Logistic Regression
Custom NumPy gradient descent with L1 regularization for automatic feature selection. It reduced 3 uninformative category weights to zero, with loss decreasing from 0.693508 to 0.677043.

Model 5

Scikit-Learn Logistic Regression
Benchmark model using Scikit-Learn's L-BFGS solver. It achieved 66.34% validation accuracy, compared with 59.85% from the custom implementation.

Grid Search Hyperparameter Tuning

Learning Rate (alpha)	Regularization (lambda)	Validation Loss	Validation Accuracy
0.01	0	0.679076	58.22%
0.01	0.01	0.679144	58.22%
0.01	0.1	0.679677	58.22%
0.01	1	0.681564	58.22%
0.1	**0**	**0.658178**	**59.69% (OPTIMAL)**
0.1	0.01	0.660821	59.50%
0.1	0.1	0.671587	58.22%
0.1	1	0.677521	58.22%
0.5	0	0.744899	52.41%
0.5	0.01	0.790122	47.67%
0.5	0.1	0.844596	46.48%
0.5	1	1.045944	46.48%

Table 3: Hyperparameter Grid Search Results on Validation Set.

Observed on 12 Combinations of Hyperparameters:

Optimal Parameters: Learning Rate alpha* = 0.10, Penalty lambda* = 0.00.

Validation Performance: Lowest Validation Loss J(theta) = 0.658178, Validation Accuracy = 59.69%.

Failure Modes: lambda = 1.00 caused severe underfitting (loss spiked to 1.0459); alpha = 0.50 caused gradient overshooting.

Training Loss Convergence

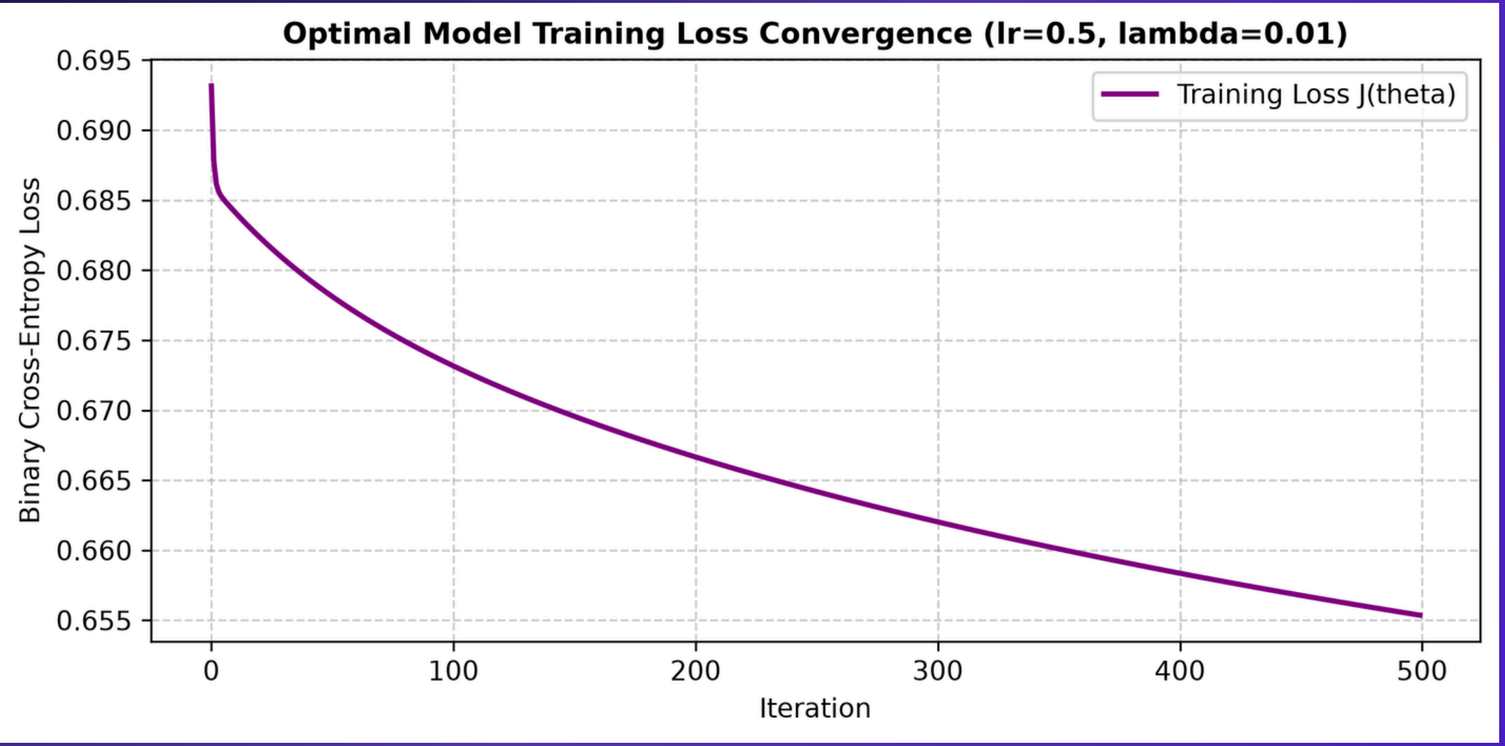


Figure 6: Optimal Model Training Loss Convergence Curve.

Threshold Calibration

Evaluation Metric	Default Threshold (T=0.50)	Calibrated Threshold (T*=0.5899)
True Positives (TP)	3,552	**2,255**
True Negatives (TN)	387	**1,760 (+1,373 gain)**
False Positives (FP)	2,352	**979 (-1,373 drop)**
False Negatives (FN)	266	**1,563**
Accuracy	60.07%	**61.23% (+1.16%)**
Precision	60.16%	**69.73% (+9.57% boost)**
Specificity (TNR)	14.13%	**64.26% (+50.13% boost)**
False Positive Rate (FPR)	85.87%	**35.74% (-50.13% drop)**
Balanced Error Rate (BER)	46.42%	**38.34% (-8.08% improvement)**

Table 4: Test Performance Comparison Across Evaluation Metrics (Default vs. Calibrated Threshold).

Confusion Matrix

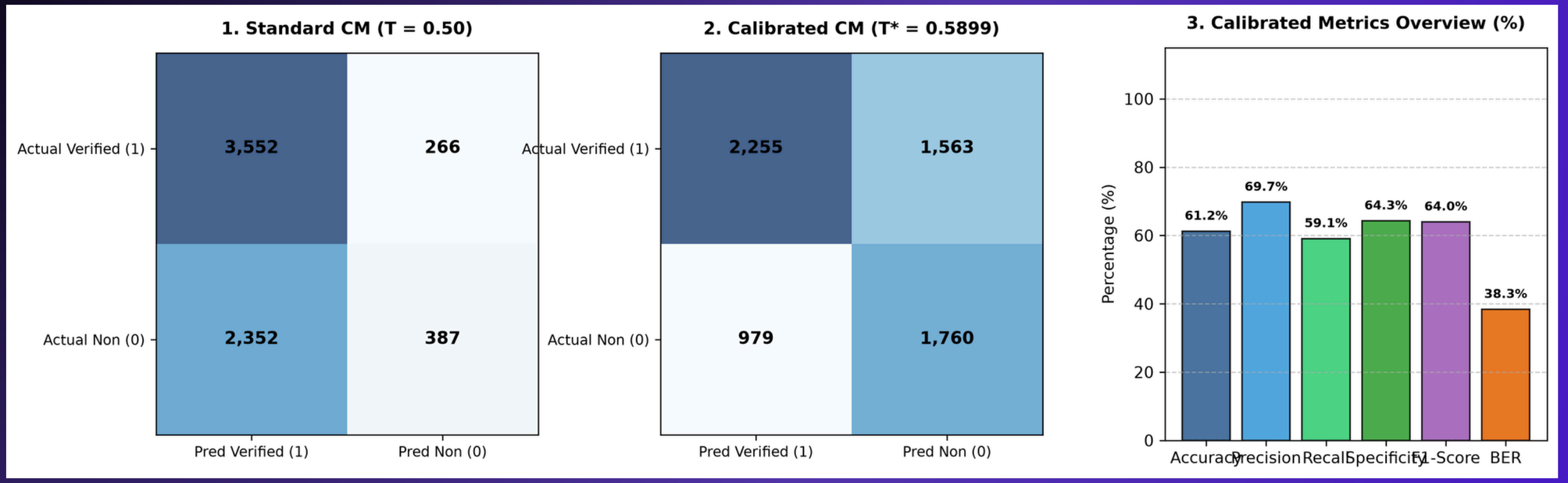


Figure 7: Test Confusion Matrices (Default 0.50 vs. Calibrated 0.5899) and Performance Bar Chart.

->The Failure of Standard Threshold (T = 0.50):
Raw accuracy looks okay (60.07%), but Specificity was only 14.13% (failed to catch non-verified reviews; BER = 46.42%).

->The Solution — Calibrated Threshold (T* = 0.5899):
Calibrated decision threshold to the validation median probability (T* = 0.5899).

->Results Breakthrough at T* = 0.5899:
• Specificity (TNR): Jumped from 14.13% to 64.26% (+50.13% massive boost!).
• Precision: Increased to 69.73% (+9.57% boost).
• Balanced Error Rate (BER): Dropped from 46.42% to 38.34%.

Limitations

Verified and non-verified reviews share overlapping feature distributions

Logistic regression assumes a linear decision boundary

Helpfulness votes are extremely sparse

Feature pipeline extracts surface-level text metrics

Reviews are classified independently

Conclusion

Behavioral and linguistic signals can automatically evaluate review trustworthiness on Rokomari.com.

Removing 538,386 duplicates and filtering upper-tail outliers reduced sample variance by 98.6%.

Statistically proved that behavioral separation exists between reviewer groups.

Built 5 classification models; minimized BCE Loss $J(\theta)J(\theta)$ via gradient descent; tuned hyperparameters ($\alpha*=0.10,\lambda*=0.00\alpha*=0.10,\lambda*=0.00$).

Calibrated decision threshold boosted Specificity to 64.26%64.26% and Precision to 69.73%69.73%, establishing an automated trustworthiness filter for Rokomari.com.

Thank You